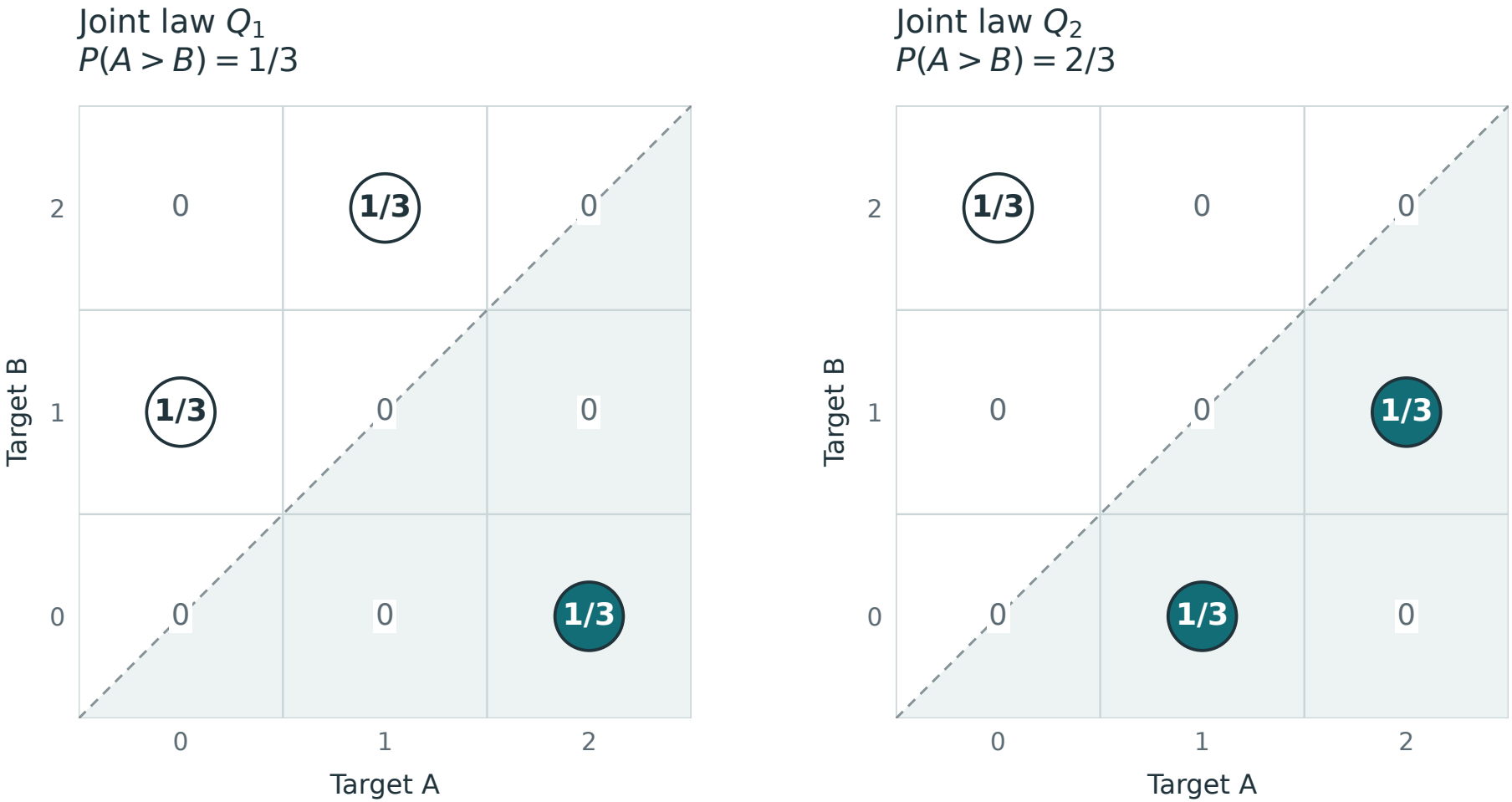
Equal marginals, different comparisons

Same marginals: A and B are each uniform on {0, 1, 2}.

Each occupied cell has probability 1/3; all remaining cells have probability 0.



Each column and each row sums to 1/3 under both laws. Only the coupling changes.

Shaded half-plane and filled supports: A > B. Dashed diagonal: A = B.

These are stipulated joint laws. Independence would supply a third law.